

159.735 Assignment 3

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Parallel Solution of the Heat Distribution Problem

1. Platform

Machine	Apple MacBook, Apple M5 system on chip
Cores	10 physical cores: 4 performance (“P”) + 6 efficiency (“E”), no SMT
Memory	32 GB unified
Caches	128 byte line; 16 MB L2 shared by the performance cluster
OS	macOS 26.3, arm64
Compiler	Apple clang 21.0.0, <code>-O3</code>
OpenMP	LLVM <code>libomp</code> (Homebrew), <code>-Xpreprocessor -fopenmp -lomp</code>
FITS	<code>cfitsio</code> 4.7.0 (Homebrew)

The heterogeneous core layout matters throughout this report. The machine reports ten cores, but they are not ten of the same thing: the four performance cores are several times faster than the six efficiency cores. Almost every departure from ideal scaling measured below traces back to that fact.

2. The problem and the numerical method

Laplace’s equation is solved on a square plate by explicit finite differences. The updated temperature of a pixel is the average of its four neighbours,

$$g(y,x) = (h(y,x-1) + h(y,x+1) + h(y-1,x) + h(y+1,x)) / 4$$

which is Jacobi iteration: the whole new image `g` is computed from the old image `h`, and only then does `g` become the current image.

The plate edge, the cold finger heat sink and the printed circuit components are Dirichlet boundary conditions — they hold a fixed temperature for the whole run. They are drawn by `fix_boundaries2()` from the supplied `draw.hxx`, unmodified.

Convergence. A pixel has settled when it moved by less than `tol = 1e-5 K` in a sweep. The plate has converged when every interior pixel has settled. Pixels on the outer edge are fixed, so they are converged by definition and are left out of the count; requiring all `(npix-2) x (npix-2)` interior pixels to settle is exactly equivalent to requiring all `npix x npix` of them.

The number of iterations this takes is not known in advance. Jacobi’s convergence rate is set by the spectral radius of its iteration matrix, $\cos(\pi/(n+1)) \sim 1 - \pi^2/2n^2$, which predicts an iteration count growing as n^2 . The measured counts fit $n^{1.83}$ – slightly below n^2 because the stopping test is an absolute threshold on the per-iteration *change*, and the change at a given error level is itself proportional to $1/n^2$, so the test is relatively easier to satisfy on a larger plate. Folding that in, the model $k = A n^2 (B - 2 \ln n)$ reproduces all four measured counts to within 2.5%:

plate	100x100	200x200	400x400	800x800
iterations to converge	3,711	13,543	47,875	165,265

3. The programs

File	What it is
heat.cpp	sequential solution
heat_omp.cpp	OpenMP parallel solution
heatutil.hxx	everything the two have in common — the Jacobi sweep itself, the fixed-pixel mask, the row partitioning, the timer, the image hash
makefile	make heat, make heat_omp, or just make
run_scaling.sh, analyse.py	the measurements and figures in this report
make_plate_figure.py	renders the FITS output as a picture

array.hxx, arrayff.hxx, fits.hxx, fitsfile.h, fitsfile.cpp and draw.hxx are used exactly as supplied.

Putting the kernel in a shared header is not tidiness for its own sake. The sequential and parallel programs call **the same jacobi_sweep() function**, so a given pixel goes through the identical arithmetic in the identical order whichever program computes it. That is the mechanism behind the bit-identical results in section 5 — it is a structural property, not something that happens to hold.

```
make
./heat 800                # sequential, writes plate0.fit and plate1.fit
./heat_omp 800 -p 8        # 8 threads, writes plate0.fit and plate1_omp.fit
./heat_omp 800 -p 8 -d     # dynamic partition instead of static blocks
```

Options: -p N threads, -t TOL tolerance, -o FILE output (none to skip file writing when timing), -i N stop after N iterations (a benchmarking aid only), -d dynamic partition.

One optimisation worth describing

The obvious way to hold the circuit at a fixed temperature is to call `fix_boundaries2()` on the new image every iteration, re-drawing the whole circuit 165,000 times. Instead the program works out **once**, before the loop, which pixels the drawing routines touch. It does this by running the supplied drawing code on a scratch array pre-filled with a sentinel value; any pixel that is no longer the sentinel afterwards is a fixed pixel. The sweep then simply carries those pixels through unchanged.

This is numerically identical — the same pixels end up with the same values — but it removes a serial $O(n^2)$ section from the middle of the iteration loop, which would otherwise have put a hard floor under the parallel runtime.

Both programs also **swap the two image buffers** rather than copying `g` over `h`, which removes n^2 float copies per iteration.

4. The parallel implementation

4.1 Partitioning

The plate is divided into horizontal strips of whole rows. Rows are the unit of work because a row is contiguous in memory, so a thread that owns rows `[y0, y1)` reads only **two** rows it does not own: the halo row immediately above its strip and the halo row immediately below it.

Two ways of handing out the rows are implemented:

Static blocks (default). Each thread gets one contiguous block, assigned once before the loop starts. The remainder is spread one row at a time over the first few threads, so no two threads ever differ by more than a single row. There is no bookkeeping during the run, and cache locality is as good as it can be because a thread revisits its own rows every iteration.

Dynamic self-scheduling (`-d`). The rows are cut into chunks of about `nrows / (4 * nthreads)` and threads take the next chunk from a shared counter via `#pragma omp atomic capture`. This costs one atomic increment per chunk and gives up some locality, but a slow thread simply takes fewer chunks. On a machine with four fast cores and six slow ones, that turns out to matter a great deal (section 7.3).

Both were implemented because the textbook answer — a static block partition — is provably the right choice on a homogeneous machine and demonstrably not the right choice on this one.

4.2 The thread pool

The pool is created **once**, outside the iteration loop:

```
#pragma omp parallel default(shared)
{
    ...
    for (;;) {           // the whole do-while loop lives inside the region
        ...
    }
}
```

The alternative — a `#pragma omp parallel for` inside the loop — would fork and join a team 165,000 times for an 800x800 plate.

4.3 Synchronisation

This is the local synchronisation problem from the lectures: a thread must not run ahead of the neighbours it shares halo rows with. Two team-wide barriers per iteration are enough, and are far simpler than pairwise neighbour signalling:

```
counts[tid].n = jacobi_sweep(h.buffer, g.buffer, fixedp, npixx, y0, y1, tol);

#pragma omp barrier                                // (1)

#pragma omp single
{
    int total = 0;
    for (int t = 0; t < nth; ++t) total += counts[t].n;
    ++iter;
    done = (total >= nrequired) || (iter >= itmax);
    std::swap(h.buffer, g.buffer);
}                                                    // (2) implicit barrier

if (done) break;
```

Barrier (1) is the important one. Without it a fast thread could finish its strip, start the next iteration, and begin overwriting a row of `h` that a neighbour is still reading as a halo row. It also guarantees every per-thread convergence count is visible before anyone reads them.

Barrier (2), implicit at the end of `single`, protects the buffer swap: while one thread swaps the pointers, every other thread is parked at that barrier and cannot be reading either image. After it, all threads see the swapped buffers and the same termination verdict.

A team-wide barrier is stronger than this problem strictly needs — a thread only genuinely depends on its two neighbours — but `libomp`'s barrier is a tuned, actively-spinning tree barrier, and hand-rolled neighbour flags would have to reimplement that badly. Section 7.2 measures what it costs.

4.4 Detecting convergence and terminating cleanly

The iteration count cannot be fixed in advance, so convergence has to be tested every iteration, and every thread has to agree on the answer.

Each thread counts settled pixels **in its own rows only**, and writes that count into its own slot of a shared array. Each slot occupies a full 128 byte cache line:

```
struct PaddedCount { int n; char pad[CACHE_LINE - sizeof(int)]; };
```

Packed together, ten counters would share one or two cache lines and the threads would spend their time bouncing those lines between cores — false sharing, and on a per-iteration counter it is expensive.

After barrier (1), one thread sums the slots and compares against $(npix-2)^2$. The verdict goes into **one shared bool done**, and every thread reads that same flag after barrier (2).

That single shared flag is what makes termination clean. Every thread evaluates `if (done) break;` on the same iteration, having read the same value after the same barrier. So either the whole team continues or the whole team leaves. No thread can exit early and leave a neighbour's halo row unwritten, and — the failure mode that actually bites in practice — no thread can be left waiting forever at a barrier that the rest of the team has already walked away from.

4.5 Why the answer does not depend on the thread count

Three properties combine:

1. **Jacobi is partition-independent.** A new pixel value is a function of the *previous* image only. Splitting the rows between threads changes who computes a pixel, never what it computes.
2. **Both programs run the same code.** `jacobi_sweep()` is shared, so the four additions and the multiply happen in the same order, with the same float rounding, in both programs.
3. **The convergence test is exact.** It is a count of integers reduced by addition — no floating point rounding is involved in the stop decision, so the stop happens on exactly the same iteration every time.

Consequently the parallel program does not merely produce a *similar* image in a *similar* number of iterations; it produces the identical image in the identical number of iterations, which is what section 5 verifies.

5. Verification

Each program prints an FNV-1a hash of the raw bytes of the final image. This is compared rather than the FITS files themselves, whose headers carry timestamps.

Every plate size was run to full convergence with the sequential program and with the OpenMP program at a range of thread counts, in both partition modes:

npix	iterations	distinct iteration counts	distinct image hashes	runs compared
100	3,711	1	1	30
200	13,543	1	1	30
400	47,875	1	1	30
800	165,265	1	1	13

One iteration count and one image hash per plate size, across every configuration tested. The requirement that the parallel program run in the same number of iterations as the sequential version and produce the same output is met exactly, not approximately.

6. How the timings were taken

Two measurement problems had to be dealt with first, and both are worth recording because both produced confidently wrong numbers before they were noticed.

`omp_get_wtime()` counts time the machine spends asleep. LLVM's runtime implements it with `gettimeofday()`, which is wall clock time. This laptop is configured to sleep after one minute idle on battery, so an unattended benchmark reported runs of up to 4,893 seconds inside a suite that used about fifteen minutes of CPU — the sleep was being billed to whichever run happened to straddle it. Worse, the sequential program was using `steady_clock` while the parallel program used `omp_get_wtime()`, so the two were not even being timed by the same clock. Both programs now share one `wall_seconds()` helper built on the monotonic `steady_clock`, which does not advance across a system sleep, and the benchmark script re-executes itself under `caffeinate` so the machine cannot doze off mid-run.

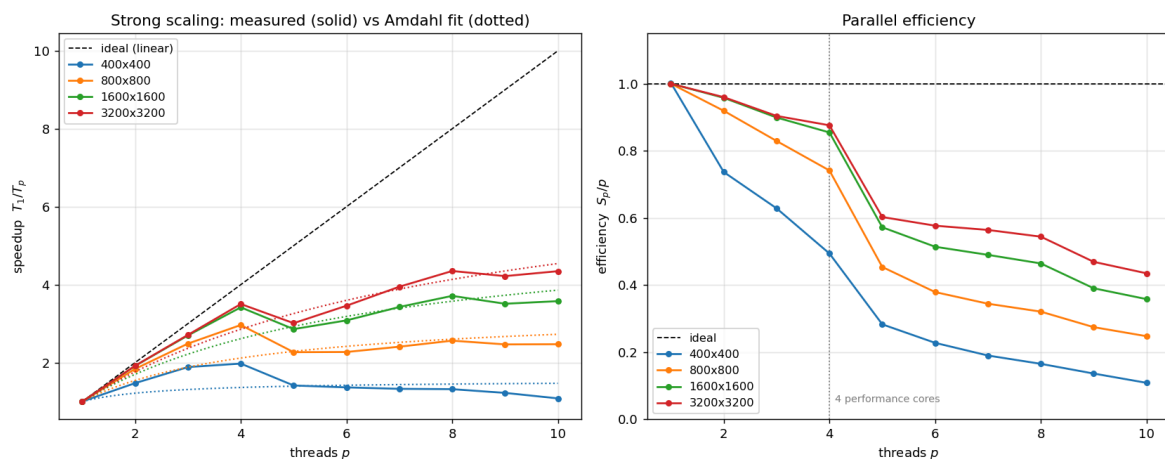
The machine has other work to do. This is a laptop running a browser, an IDE and an office suite; the load average sat between 3 and 6.6 during the measurements. Each configuration is therefore measured five times with the repeats interleaved, and the **fastest** run is used, that being the reading least polluted by other processes. The load average is recorded alongside every measurement. Across all configurations the median run was 1% slower than the fastest and the worst was 11% slower, so the numbers below are repeatable to a few percent — good enough for the conclusions drawn, though not laboratory-grade.

Timing runs use a fixed iteration budget (`-i`) chosen so that a sequential run takes about a second. This only bounds how long a measurement takes; the work per iteration is identical for every thread count, so the speedups are unaffected. The runs in section 5 and the picture in section 8 are complete runs to convergence.

7. Results

7.1 Strong scaling — Amdahl's law

Fixed plate, varying thread count. Speedup is measured against the **sequential program**, not against the parallel program on one thread.



strong scaling

plate	best speedup	at threads	fitted Amdahl f	ceiling 1/f
400x400	1.97x	4	0.646	1.5x
800x800	2.96x	4	0.296	3.4x
1600x1600	3.71x (4.23x dynamic)	8–9	0.177	5.7x
3200x3200	4.35x (4.85x dynamic)	8–10	0.134	7.5x

Selected detail for the 3200x3200 plate:

threads	time (s)	speedup	efficiency	thread busy %
1	0.843	1.00	1.00	100%
2	0.439	1.92	0.96	98%
3	0.311	2.71	0.90	98%
4	0.241	3.50	0.88	96%
5	0.280	3.01	0.60	73%
8	0.194	4.35	0.54	77%
10	0.194	4.35	0.43	69%

Does it obey Amdahl's law? Yes, in the sense that matters: the speedup saturates rather than growing with p , and the measured curve is described well by $S(p) = 1 / (f + (1-f)/p)$. The fitted serial fraction falls steadily as the plate grows — 0.646, 0.296, 0.177, 0.134 — which is the central Amdahl prediction: **the same program has a different ceiling depending on how much work each processor is given**. At 400x400 the ceiling is about 1.5x no matter how many cores are thrown at it, and the measurements duly stop at 1.97x; at 3200x3200 the ceiling is about 7.5x.

Two honest qualifications:

- The f fitted here is not literally serial code. The only genuinely serial region in the program is the **single block**, which adds at most ten integers. What f actually captures is **per-iteration overhead and load imbalance**, which happen to have the same functional effect on the speedup curve. Amdahl's law is being used here as a phenomenological description, which is how it is normally used in practice, but it is worth being precise about.
- The curve is not smooth. There is a pronounced **break between 4 and 5 threads** in every single measurement, and efficiency drops off a cliff there — from 88% to 60% at 3200x3200. That is not noise and it is not Amdahl; it is the fifth thread landing on an efficiency core. Section 7.3.

At the small end, the parallel program is simply **slower** than the sequential one:

npix	iterations	sequential (s)	best parallel (s)	threads	speedup
100	3,711	0.016	0.016	1	1.00
200	13,543	0.243	0.242	1	1.01
400	47,875	3.580	1.884	4	1.90
800	165,265	51.883	17.711	4	2.93

For a 100x100 plate, running on ten threads takes **almost thirteen times longer** (0.200 s against 0.016 s) than running sequentially. Section 7.2 explains why, and it is the purest possible illustration of Amdahl's

point: below a certain problem size, parallelism costs more than it buys.

7.2 Where the overhead comes from

A 64x64 plate is small enough that the arithmetic is nearly free, so whatever the parallel run costs above the sequential one is the price of the two barriers and the reduction:

threads	2	4	6	8	10
overhead (us/iteration)	8.1	14.7	32.7	41.7	59.0

Roughly 6 microseconds per additional thread. Compare that with the sequential cost of one iteration:

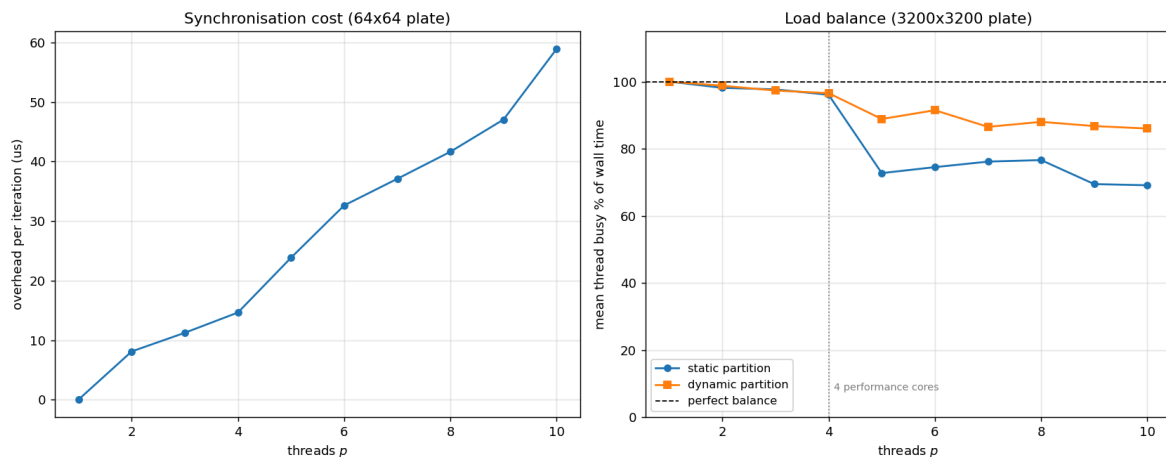
plate	400x400	800x800	1600x1600	3200x3200
sequential time per iteration	76 us	312 us	1,340 us	5,440 us
10-thread overhead as a fraction of it	78%	19%	4.4%	1.1%

This is the whole story of the size dependence in one line. At 400x400 the synchronisation needed to run an iteration on ten threads costs nearly as much as computing the iteration outright, so ten threads cannot win. At 3200x3200 it is one percent and effectively free.

Note also that the overhead grows with thread count while the work per thread shrinks — the two move in opposite directions, which is why every curve in the speedup plot eventually turns over.

7.3 Load balance and the two kinds of core

Each thread accumulates how long it spent inside `jacobi_sweep()`. Comparing that against the wall clock says how much of the run a thread spent working rather than waiting at a barrier.



overhead and load balance

3200x3200 plate, mean thread busy percentage:

threads	2	3	4	5	6	8	10
static partition	98%	98%	96%	73%	75%	77%	69%
dynamic partition	99%	97%	97%	89%	91%	88%	86%

The static partition holds 96–99% efficiency for up to four threads and then falls off a cliff at the fifth. The reason is that the static partition gives every thread an equal number of rows, which is the correct thing to do only if every core is equally fast. The fifth thread runs on an efficiency core, takes far longer over its equal share, and **every barrier runs at the speed of the slowest thread**. At ten threads the busiest thread is working 85% of the time while the idlest works 48% — half the pool is sitting at a barrier.

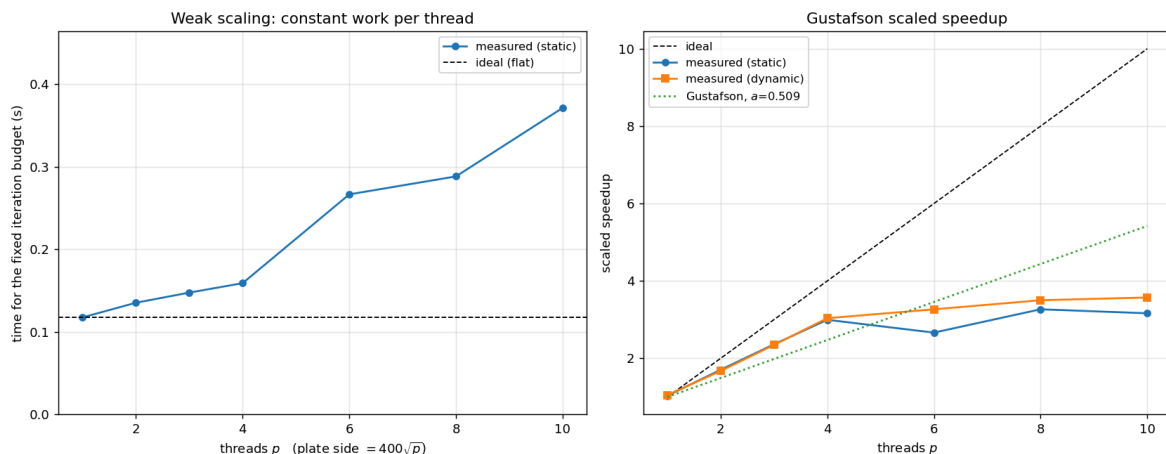
The dynamic partition fixes most of this without any knowledge of which core is which: a thread on a slow core simply claims fewer chunks. It lifts balance at ten threads from 69% to 86% and the best speedup from 4.35x to 4.85x. It costs a little at four threads, where the static partition's perfect locality wins and there is no imbalance to correct.

The size of the effect can be estimated from the data. Four performance cores deliver 3.50x. Adding six efficiency cores takes that to 4.85x, so the six of them together contribute about 1.35x — roughly **a quarter of a performance core each** for this memory-bound kernel. Ten cores, but only about five and a half cores' worth of throughput, which caps any speedup on this machine at around 5x however good the software is.

Memory bandwidth is the other ceiling at the largest sizes. At 3200x3200 the two images and the mask occupy 92 MB, far beyond any cache, so the kernel streams from DRAM. Taking about 9 bytes of traffic per pixel update (one new float read, one float written, one mask byte — the three active rows stay resident), the sequential run moves about 17 GB/s and the best 10-thread run about 82 GB/s, against roughly 150 GB/s of peak bandwidth on this chip. At better than half of peak the memory system is a real co-limiter, and no amount of load balancing would remove it.

7.4 Weak scaling — Gustafson's law

The plate side grows as $400 \cdot \sqrt{p}$, so the number of pixels per thread — and therefore the arithmetic each thread does per iteration — stays constant at about 160,000. Perfect scaling keeps the parallel time flat and gives a scaled speedup of p .



weak scaling

p	npix	pixels/thread	sequential (s)	parallel (s)	scaled speedup	dynamic	scaled speedup
1	400	160,000	0.121	0.118	1.03	0.116	1.05
2	566	160,178	0.230	0.135	1.70	0.138	1.67
3	693	160,083	0.348	0.147	2.36	0.149	2.35
4	800	160,000	0.476	0.159	3.00	0.157	3.04
6	980	160,066	0.710	0.266	2.66	0.217	3.27
8	1131	159,895	0.941	0.288	3.27	0.269	3.50
10	1265	160,022	1.174	0.371	3.16	0.329	3.57

Fitting $S = p - a(p-1)$ gives $a = 0.51$ static, $a = 0.48$ dynamic.

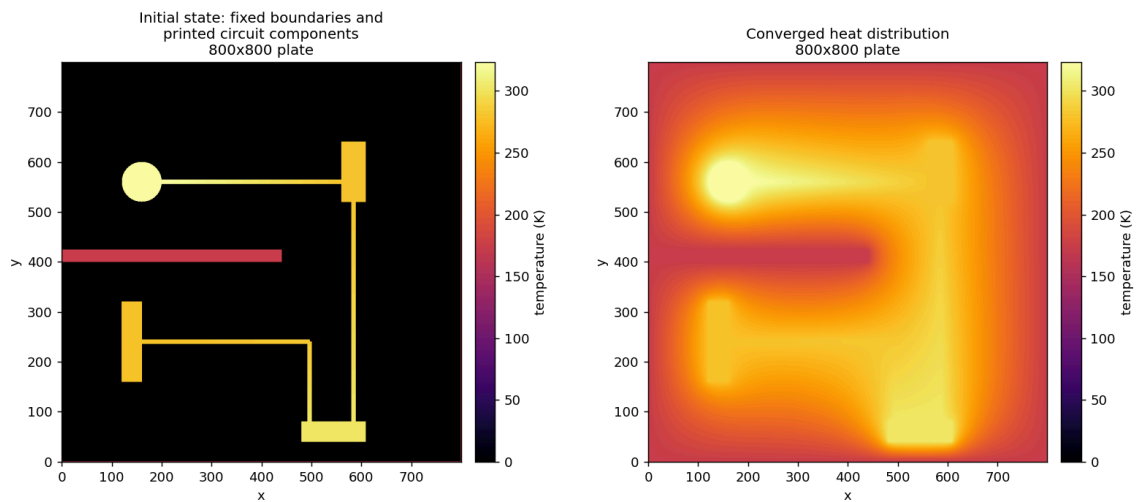
Does it scale up for larger problems in accordance with Gustafson's Law? Partly, and the way it fails is informative.

Gustafson's point is that the *sequential* time grows with the scaled problem while the parallel time need not — and that is exactly what the third and fourth columns show. Sequential time rises almost perfectly linearly, 0.121 s to 1.174 s, a factor of 9.7 for a factor of 10 in work. Meanwhile parallel time rises only from 0.118 s to 0.329 s. Ten times the problem is solved in under three times the time. That is the Gustafson argument working: **enlarging the problem in step with the machine is far more effective than trying to make a fixed problem faster**. Compare the 3.57x scaled speedup at ten threads against the 1.08x that ten threads managed on a fixed 400x400 plate in section 7.1 — the same program, the same threads, an entirely different outcome.

But it does not reach the ideal $S = p$ line, and it stops improving after four threads. Up to four threads the scaled speedup tracks the ideal closely (1.70, 2.36, 3.00 against 2, 3, 4); beyond that it flattens. The cause is the same as before: threads five to ten are on efficiency cores, so “constant work per thread” is not constant work per unit time. Gustafson's law assumes p equal processors, and this machine does not have ten of those. On the four performance cores alone the law is obeyed almost exactly; the fitted $a = 0.5$ is measuring hardware heterogeneity, not a serial section in the program.

8. The heat distribution

800x800 plate, run to convergence — 165,265 iterations, tolerance 1e-5 K. Final temperatures range from 173 K (the cold plate edge and the cold finger) to 323 K (the hot circular component), mean 233.21 K.



heat distribution

The left panel is the initial state: the fixed boundary, the cold finger heat sink as the dark horizontal bar, and the printed circuit components, with the rest of the plate at 0 K. The right panel is the converged solution. Heat has spread from the components into the plate, the cold finger has cut a distinct cool channel across the middle, and the cold edge draws heat out all round. The isotherms bulge away from the hot circular pad at upper left and are compressed between the components and the cold finger, which is where the steepest gradients — and in a real board, the thermal stress — would be.

9. Conclusions

1. The OpenMP program produces a **bit-identical image in an identical number of iterations** to the sequential program, at every thread count and both partition strategies. This follows structurally from Jacobi being partition-independent, the two programs sharing one kernel, and the convergence test being an exact integer count.
2. **Amdahl's law is obeyed**, with the fitted serial fraction falling from 0.646 at 400x400 to 0.134 at 3200x3200. The per-iteration synchronisation cost — measured directly at 8 us for two threads rising to 59 us for ten — is what sets that fraction, and it explains both the ceiling at each size and why a 100x100 plate runs nearly thirteen times slower on ten threads than on one.
3. **Gustafson's law is obeyed on the performance cores and limited by hardware beyond them.** Ten times the problem is solved in 2.8 times the time, and scaled speedup tracks the ideal line closely up to four threads before flattening at about 3.6.
4. **The binding constraint on this machine is that it does not have ten equal cores.** Four performance cores give 3.50x at 88% efficiency; the six efficiency cores add only about 1.35x between them. A static equal-row partition, which is the textbook-correct choice, is actively the wrong one here because every barrier waits for the slowest thread — measured busy time falls to 69%. Switching to dynamic self-scheduling recovers balance to 86% and lifts the best speedup from 4.35x to 4.85x, without changing a single output value.
5. If more speedup were needed, the next steps would be algorithmic rather than mechanical: red-black Gauss-Seidel or successive over-relaxation would cut the *number* of iterations by a large factor, which is worth far more than the remaining few tens of percent available from tuning the parallel

decomposition. Multigrid would be better still. Jacobi's n^2 iteration count, not its per-iteration parallelism, is the real cost of this method.